

FINAL YEAR RESEARCH PROJECT PROPOSAL

1. Research Topics

1.1 Primary Research Direction

- Multi-Agent Strategy Orchestration on the Colombo Stock Exchange under Dynamic Volatility Memory and Low Liquidity Friction (i.e. real market conditions)
 - **Task:** Multi-Agent Strategy Orchestration
 - **Domain:** Colombo Stock Exchange (Frontier Equity Market)
 - **Constraint:** Dynamic Volatility Memory and Low-Liquidity Execution Friction

1.2 Alternative Research Direction

- Multi-Horizon Sequential Price Forecasting on the Colombo Stock Exchange via Explainable Localized Multilingual Sentiment Fusion
 - **Task:** Multi-Horizon Sequential Price Forecasting
 - **Domain:** Colombo Stock Exchange (Frontier Equity Market)
 - **Constraint:** Explainable Localized Multilingual Sentiment Fusion

2. Problem Background and Context

The Colombo Stock Exchange (CSE) represents a classic frontier equity market characterized by low retail participation, high systemic volatility, and pronounced informational asymmetry. Active capital market participation in Sri Lanka is restricted to approximately 1% of the population of 22 million. Out of 850,000 registered Central Depository System portfolios, only 200,000 are active, with retail investors frequently exiting due to losses and cognitive overload caused by highly technical corporate financial reports.

Unlike deep, liquid Western exchanges, the CSE is a retail-dominated, low-liquidity market prone to sudden price shocks and transaction-induced slippage. The boundaries of this project encompass a simulated multi-agent trading framework modeled using LangGraph(TBD) to evaluate trading strategies, portfolio optimization, and risk mitigation policies under realistic liquidity constraints.

3. Literature Review Matrix

Reference Study	Core Target Task	Underlying Dataset	Extracted Feature Set	Implemented Model / Architecture	Evaluation Metrics	Documented Limitations
B. S. Senarathna et al. (2025)	Computerized algorithmic trading and trend forecasting.	20-year daily historical stock prices of 4 S&P SL20 companies.	Daily closing prices and transformer-extracted text sentiments.	Long Short-Term Memory Network integrated with technical indicators.	Cumulative Returns, Sharpe Ratio, Maximum Drawdown.	Limited to 4 specific high-cap stocks; does not address transaction costs or order book slippage.
Dissanayake & Nanayakkara (2021)	Return forecasting under sentiment and liquidity shifts.	Monthly CSE historical database spanning 2010 to 2021.	Market-wide turnover ratios and calculated bid-ask illiquidity.	Ordinary Least Squares linear time-series regression.	Adjusted R^2 , Coefficient t-statistics, F-significance.	Assumes strictly linear relationships; fails to capture intraday non-linear volatility regimes.
A. Rizvi et al. (2025)	Multilingual sentiment monitoring for financial reviews.	13,000 aspect-tagged local banking customer comments.	Morphological text, Singlish, and Sinhala-English code-mixed tokens.	Hybrid XLM-RoBERTa with low-rank adaptation.	Classification Accuracy, Macro-F1, SHAP/LIME feature weights.	Focused on customer service aspects; lacks integration with time-series stock price indicators.
Sangiorgi &	Multi-agent	TradingAgents Cache	Financial news	Multi-agent LLM	Simulated execution	Limited strictly to

Wharton Initiative (2025)	financial framework trading and market outcome analysis.	DB containing US equity feeds.	headlines, Reddit chatter, insider trades, and US tickers.	financial orchestration graph.	returns, agent transaction logs, strategy consensus metrics.	deep, highly liquid US equities; completely lacks frontier market friction or local sentiment pipelines.
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4. Derived Research Gap

A systematic cross-paper pattern analysis isolates a critical **Execution Friction and Volatility Gap**:

Existing quantitative trading frameworks and multi-agent systems operate under the idealized assumption of frictionless, highly liquid markets. They completely ignore the structural realities of frontier markets, such as long-term volatility clustering, asynchronous trading volumes, and transaction-induced slippage. Furthermore, current sentiment models remain strictly monolingual or domain-isolated, failing to connect unstructured, low-resource multilingual (Sinhala, Tamil) and code-mixed (Singlish) retail forum chatter directly to multi-horizon predictive models.

5. Contribution to the Body of Knowledge

The primary contribution of this research is highly technical, focusing on the architectural design of a decentralized coordination graph

- **Core Technical Contribution:** Designing a decentralized Multi-Agent Critique-Correction Graph Framework (built using LangGraph TBD) that structures real-time collaborative debates between asymmetric financial intelligence nodes to derive optimal trading policies.
- **Algorithmic Integration:** Incorporating a mathematical fractional-integration volatility layer to capture long-memory asset variance alongside an intraday execution simulator that enforces localized liquidity penalties.

- **Linguistic Contribution:** Introducing an aspect-level cross-lingual financial parser that channels sparse retail forum text directly into an agent consensus pipeline.

6. Feature Set Identification (AI-Based Pipeline)

Input data streams are categorized into distinct structural layers and mathematically justified to ensure pipeline validation.

- **Microstructure Ingestion Features (Core):** Intraday bid-ask spreads, order book queues, volume-at-price ticks, and real-time transaction density profiles. RAG of financial reports.
 - **Justification:** Microstructure cues directly regulate transaction slippage parameters in illiquid order books
- **Long-Term Risk Features (Derived):** Moving Average Convergence Divergence (MACD), Relative Strength Index (RSI), rolling conditional variances, and historical maximum drawdown arrays.
 - **Justification:** Time-series derivatives map mathematical momentum and capture recurring structural shocks
- **Linguistic Sentiment Features (Auxiliary/ Contextual):** Polarity vectors and aspect embeddings extracted from financial media streams, localized subreddits, and active CSE Facebook groups.
 - **Justification:** Localized retail chatter shapes immediate market trends and drives irrational order fluctuations.
- **Agent Context Features (Static Covariates):** Organizational sector categories, asset capitalization tiers, listed security classes (voting vs. non-voting shares), and global agent memory states.
 - **Justification:** Static baselines constrain the operational boundaries of distinct multi-agent decision nodes.

7. Systematic Evaluation of Candidate Datasets

Candidate Dataset	Origin and Domain	Volume / Size	Key Features Included	Inherent Limitations	Primary Task Alignment
CSE Annual Disclosures Corpus	Listed companies on the Colombo Stock Exchange.	500+ PDF annual and quarterly reports spanning 10 years.	Narrative qualitative disclosures, audited financial tables, sector tags.	Highly variable formatting, scanned layout noise, sparse manual annotations.	Main Framework : Training data for layout-aware RAG extraction blocks.
CSE Historical OHLCV Database	Unofficial reverse-engineered CSE API endpoints.	20-year daily historical records of S&P SL20 companies.	Daily Open, High, Low, Close, Volume, and ASPI index records.	Limited intraday order queues; completely lacks historical bid-ask spreads.	Main & Backup : Feeds macroeconomic time-series layers to risk engines.
Multilingual Banking Reviews Corpus	Customer feedback from Sri Lankan commercial banks.	13,000 aspect-tagged reviews.	Multilingual text (English, Sinhala, Singlish, code-mixed formats).	Domain-restricted to retail consumer banking; lacks capital market vocabulary.	Backup Framework : Pre-training validation for low-resource financial tokenizers.
Senevirathne News	Scraped Sinhala	15,059 annotated	Pure Sinhala text, 4-class	Contains general	Backup Framework

Comments Dataset	online public media portals.	comment strings.	polarity labels (Positive, Negative, Neutral, Conflict).	news commentary; completely lacks specialized corporate investment terms.	: Validating cross-lingual translation consistency profiles.
TradingAgents Cache DB	Tauric Research public repositories .	Multi-modal global financial dataset streams.	Financial headlines, Reddit text streams, insider transactions, liquid US tickers.	Restricted entirely to high-liquidity US equities (e.g., AAPL, GOOGL).	Main Framework : Baseline benchmarking of multi-agent graph communication logic.

Final Selected Dataset Strategy

A single dataset cannot resolve our target execution and volatility gap. This project fuses the CSE Historical Database (to capture raw mathematical trends) with a custom-scraped Localized Retail Media Stream from Reddit and Facebook groups (to map behavioral market fluctuations), and Financial Reports from the CSE. The structural framework will be cross-benchmarked using the clean operational parameters of the TradingAgents Cache DB to ensure baseline validation.

8. Project Challenges and Operational Mitigations

To maintain delivery boundaries, technical risks have been anticipated and coupled with rigorous mitigation plans.

- **Risk 1:** Local Web Portal Blocking & Scraping Rate Limits.
 - **Impact:** Disrupts news scraping pipelines, resulting in incomplete textual sentiment datasets.

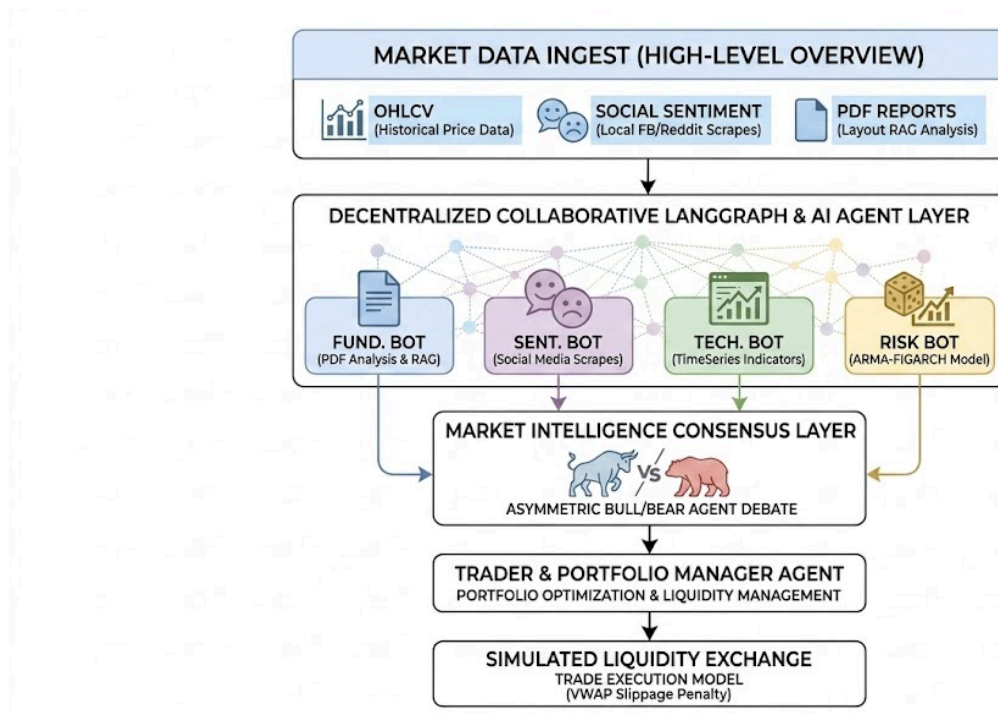
- **Mitigation:** Deploy custom API abstractions using “tvDatafeed” or cached endpoints, enforcing random sleep intervals and proxy rotations.
- **Risk 2:** Token Exhaustion and Context Chunk Loss in Long Disclosures.
 - **Impact:** Splits financial tables, isolating numeric metrics from column headers and triggering LLM hallucinations.
 - **Mitigation:** Implement layout-aware document segmentation to parse tables into structured JSON blocks before context mapping.
- **Risk 3:** Sparsity of Annotated Local Financial Texts.
 - **Impact:** Induces extreme model overfitting during aspect-level sentiment classification.
 - **Mitigation:** Leverage cross-lingual knowledge distillation from high-resource English models to low-resource targets with target consistency filters.
- **Risk 4:** High API Consumption Costs and Iterative Processing Latency.
 - **Impact:** Delays multi-agent communication rounds and boundaries due to expensive token fees.
 - **Mitigation:** Deploy lightweight, 4-bit quantized local models (such as Qwen2.5-Code or LLaMA-3) hosted via local edge execution tools.

9. High Level Solution Direction

The proposed system moves away from a single-model decision pipeline, instead constructing an event-driven multi-agent graph network that replicates the decision framework of an institutional trading firm. The framework coordinates **five specialized agents** operating in a collaborative loop:

1. **The Fundamental Analyst Agent (RAG Bot):** Utilizes a Retrieval-Augmented Generation pipeline backed by a Large Language Model to ingest audited corporate PDFs, automatically extracting balance sheet metrics, calculating financial health ratios, and summarizing corporate disclosures.
2. **The Sentiment Analyst Agent (ML Model):** Scrapes real-time retail investor sentiment from localized channels, specifically targeting public financial news streams, Reddit communities, and local CSE Facebook groups. It feeds this raw text into a specialized cross-lingual tokenizer to quantify market mood.
3. **The Technical Analyst Agent:** Processes raw historical time-series indicators, computing directional price momentum and identifying reversal signals using engineered moving averages (SMA/EMA) and the Relative Strength Index (RSI).

4. **The Volatility Risk Analyst Agent (Debating whether to implement this):** Acts as a mathematical guardrail, implementing an ARMA-FIGARCH time-series model to dynamically assess frontier market volatility persistence and calculate downside risk thresholds.
5. **The Trader & Portfolio Manager Agent:** The central orchestrator that manages a global state memory. It ingests the conflicting recommendations from the other agents, coordinates a structured automated critique/debate between bullish and bearish nodes, optimizes asset allocation, and schedules orders against simulated volume-weighted execution constraints.



References

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